Trade and Tariffs

Pathways to Arts & Humanities Summer Scholars Program

Instructor: Tudor Schlanger

Summer 2026

Review: Tragedy of the Commons

Question

A town has a public lake where anyone can fish for free. Over time, the fish population declines sharply.

Which combination of properties makes the fish a **common resource**?

A. Rivalrous and excludable
B. Non-rivalrous and non-excludable
C. Rivalrous and non-excludable
D. Non-rivalrous and excludable

Review: Tragedy of the Commons — Answer

Answer

The correct answer is **C. Rivalrous and non-excludable**.

The fish are **rivalrous**: one person's catch reduces what is available for others.

The lake is **non-excludable**: anyone can fish without paying.

This combination creates the Tragedy of the Commons — overfishing depletes the resource.

Claim 1

“International trade is generally good for a country’s economy.”

⇑ Agree

⇓ Disagree

? Not sure

Claim 2

“International trade is generally good for <u>every sector</u> of a country’s economy.”

⇑ **Agree**

⇓ **Disagree**

? **Not sure**

Claim 3

"If international trade is not good for every sector, <u>tariffs</u> can protect those affected sectors."

$\Uparrow$	$\Downarrow$	?
Agree	**Disagree**	**Not sure**

We'll revisit these at the end of class.

Outline

1. Is Trade Good for a Country?

2. Does Everyone Benefit?

3. Are Tariffs the Answer?

4. Recap

What Is International Trade?

International Trade

The exchange of goods and services across national borders.

Imports = goods coming IN to a country *(your phone, made in China, imported to the U.S.)*

Exports = goods going OUT of a country *(U.S. soybeans, shipped to China)*

Why Do Countries Trade?

Not every country can make everything well, it at all:

- Different climates, resources, and skills
- Some countries have more workers; others have more technology
- Making everything yourself is expensive and slow
- Sometimes, it's just not possible

Coffee plantation in Colombia

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Meta headquarters in California

A Tale of Two Countries

Question

Imagine a country that *is* good at everything. Should it trade at all?

A Tale of Two Countries

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The setup: Each country has 100 workers.

United States

Good at producing both software AND coffee

Especially productive at software

Colombia

Less productive at both

But relatively better at coffee

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With 100 workers, each can produce:

	Software (units)	**Coffee** (tons)
U.S.	40	20
Colombia	10	20

The Key: Opportunity Cost

Opportunity Cost (OC)

The **value** of the **next best alternative** when you make a decision.

With 100 workers, each can produce:

	Software (units)	**Coffee** (tons)	**OC of Coffee** (software units)	**Benefit/OC of Coffee** (units/ton)
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U.S.	40	20	40	
Colombia	10	20	10	

If the U.S. shifts all 100 workers to coffee, it gives up **40 software units** to get 20 tons of coffee. So the OC of coffee = 40 software units. Similarly, Colombia gives up **10 software units**.

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	Software (units)	**Coffee** (tons)	**OC of Coffee** (software units)	**Benefit/OC of Coffee** (units/ton)
U.S.	40	20	40	1/2
Colombia	10	20	10	2

Benefit/OC = Coffee / OC of Coffee. Colombia gets **2 tons per software unit** sacrificed, while the U.S. gets only $\frac{1}{2}$. Coffee is a much better deal for Colombia!

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> The Benefit / Opportunity Cost ratio of producing Coffee is much higher for Colombia than it is for the US!

Absolute Advantage

Absolute Advantage

Being the **most productive** at producing a good—making more of it with the same resources.

	Absolute Advantage (most productive)
United States	Software & Coffee
Colombia	—

The U.S. can produce **at least as many** goods with the same number of workers. So should it even bother trading?

Comparative Advantage

Comparative Advantage

Being **relatively better** at producing something, i.e. having a higher benefit / opportunity cost ratio.

	Absolute Advantage (most productive)	**Comparative Advantage** (highest Benefit/OC)
United States	Software & Coffee	Software
Colombia	—	Coffee

The U.S. is more productive at **both** goods (absolute advantage), but each country has a comparative advantage in the good where its Benefit/OC is highest.

No Trade: Each Country Makes Everything

If each country splits its workers 50/50 between software and coffee, then the countries would produce:

	Software (units)	Coffee (tons)
United States	20	10
Colombia	5	10
World total	25	20

Not bad—but can they do better?

With Trade: Specialize and Swap

Step 1: Specialize

Each country puts all workers on its comparative advantage:

	Software (units)	**Coffee** (tons)
U.S.	40	0
Colombia	0	20

With Trade: Specialize and Swap

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	Software (units)	**Coffee** (tons)
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Step 2: Trade

The U.S. gives Colombia 10 software units for 10 tons of coffee.

	Software (units)	**Coffee** (tons)
U.S.	30	10
Colombia	10	10

Gains from Trade

No Trade (50/50 split)

	Software (units)	**Coffee** (tons)
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With Trade (specialize + swap)

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Same coffee, but **15 more software units** for the world. The U.S. gains +10, Colombia gains +5. **Both countries are better off!**

*Even though the U.S. is more productive at **both** goods, trade still helps—each country specializes in what it does relatively best.*

Claim 1: Is Trade Good for a Country?

“International trade is generally good for a country’s economy.”

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Answer

Yes—international trade generally grows the economic pie. Through specialization and comparative advantage, countries produce more together than they could alone.

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A Bigger Pie, But Who Gets the Slices?

No Trade (50/50 split)

	Software (units)	**Coffee** (tons)
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Full Specialization

	Software (units)	**Coffee** (tons)
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Assumption: Workers can be moved from one sector to another **without any cost**.

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Assumption: Workers can be moved from one sector to another **without any cost**.

Question

What is the problem with this assumption?

Costs of Specialization

Question

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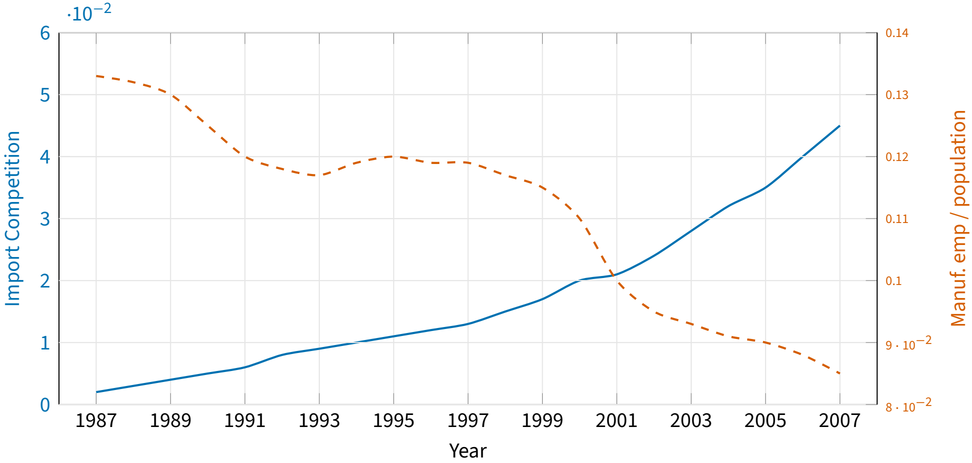
Answer

- Workers in shrinking sectors **lose their jobs** in the short run
- They need to be **retrained** for a new job, which takes time and money
- If the new job requires higher skills, they may also need a **college education**
- If the new job is in a different city, they need to be **highly mobile**, i.e. willing and able to relocate

Deindustrialization in America



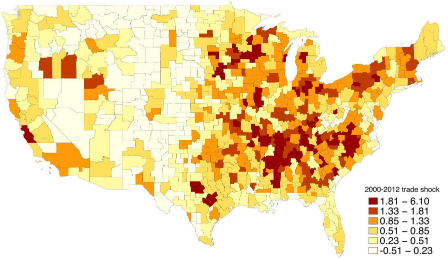
Source: Autor, Dorn & Hanson (2013), Figure 1

The China Shock

Economists studied what happened to U.S. communities most exposed to Chinese imports (Autor, Dorn & Hanson 2021 BPEA)

- China's entry to the WTO (2001) flooded the U.S. with cheap manufactured goods
- Regions most exposed saw persistent **job losses** in manufacturing, **lower income per capita**
- Net out-migration only for foreign-born workers and young native born

Figure 4: Chinese Import Penetration (2000-2012) in U.S. Commuting Zones



Source: Autor, Dorn & Hanson (2021), Figure 4

Claim 2: Does Every Sector Benefit?

"International trade is generally good for every sector of a country's economy."

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"International trade is generally good for every sector of a country's economy."

Answer

No—not every sector benefits from trade. Trade creates winners and losers. Import-competing industries shrink, and workers in those sectors face persistent job losses. Transition to growing sectors is possible in theory, but difficult, slow, and painful in practice.

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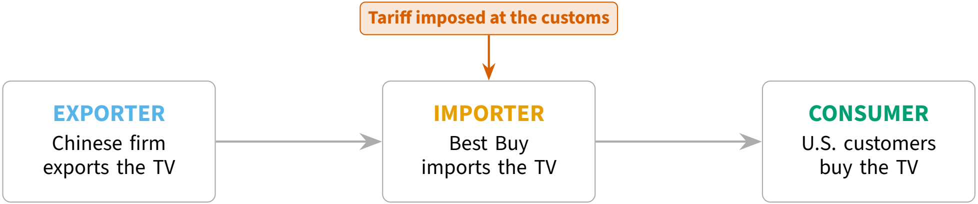
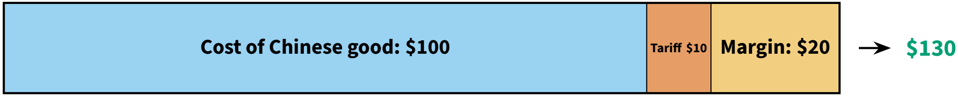


Who Bears the Cost of the Tariff?

Base case: Small tariff.



Option A — Extra tariff paid by the exporter:

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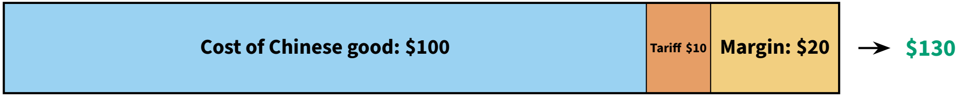
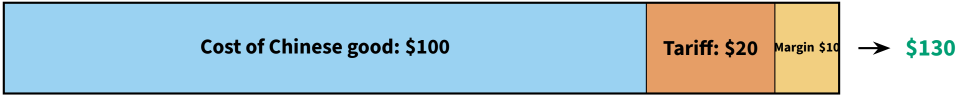


Option B — Extra tariff paid by the importer:



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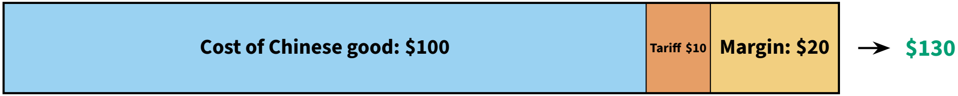

Option C — Extra tariff paid by the consumer:

Pass-Through of Tariffs

Pass-Through

The share of a tariff increase that is passed on to consumers through higher prices.

- **Option A** (**exporter** pays): pass-through = 0%
- **Option B** (**importer** pays): pass-through = 0%
- **Option C** (**consumer** pays): pass-through = 100%

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Question

What is the empirical pass-through of U.S. tariffs?

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Question

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Answer

50-100%

The Budget Lab at Yale, Jan 2026.

Why Do Countries Use Tariffs?

Economic

Protect young industries

Germany’s Zollverein (1834) shielded young industries from British competition

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EU agricultural tariffs protect French farmers despite cheaper imports abroad

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EU agricultural tariffs protect French farmers despite cheaper imports abroad

Strategic

Bargaining leverage

Use tariff threats to pressure partners into lowering their tariffs (e.g. USMCA)

Case Study: Steel Tariffs (2018)

In 2018, the U.S. government placed a **25% tariff on imported steel** and a **10% tariff on imported aluminum**.

- **What:** A tax on steel and aluminum coming into the U.S.
- **Who it applied to:** Nearly every country that sells steel to the U.S.
- **Why:** The government argued that the U.S. needs its own steel industry for **national security**

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- **Why:** The government argued that the U.S. needs its own steel industry for **national security**

Question

What are the benefits and costs?

Video: The Steel Tariff Debate

Click to watch the video

Steel Tariffs: Benefits and Costs

Arguments made in the video:

Benefits

- **Small businesses:** Small domestic steelmakers can compete better against cheap imports
- **Reciprocal trade:** China was "dumping" steel (selling below cost to destroy competitors), this tariff acts as a disincentive for steel and other goods
- **National security:** The U.S. needs to be able to make its own steel, this tariff promotes important industry

Steel Tariffs: Benefits and Costs

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Costs

- **Losses** > **Gains:** “All studies on the 2002 steel tariffs find that the costs of the tariffs outweighed their benefits in terms of GDP and employment”
- **Higher consumer prices:** Consumers end up paying more for industries relying on steel inputs
- **Wrong target:** Chinese steel is only **2%** of U.S. steel imports, so if anything, these tariffs are ineffective

Steel Tariffs: Zoom in on Employment

25% tariff on imported steel

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Domestic steel prices and employment rise
~3,500 jobs added (Read 2005)

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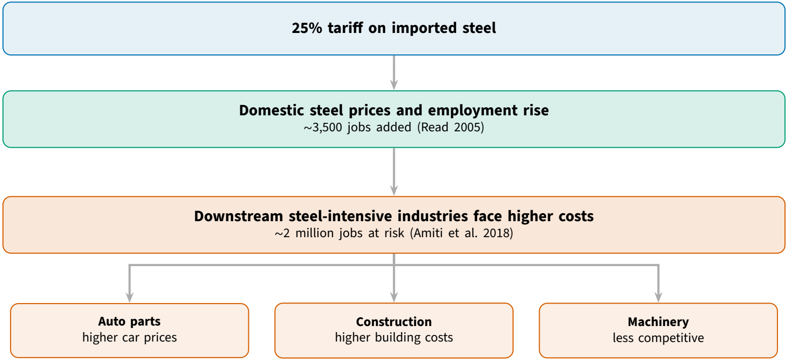
Downstream steel-intensive industries face higher costs
~2 million jobs at risk (Amiti et al. 2018)

Steel Tariffs: Zoom in on Employment



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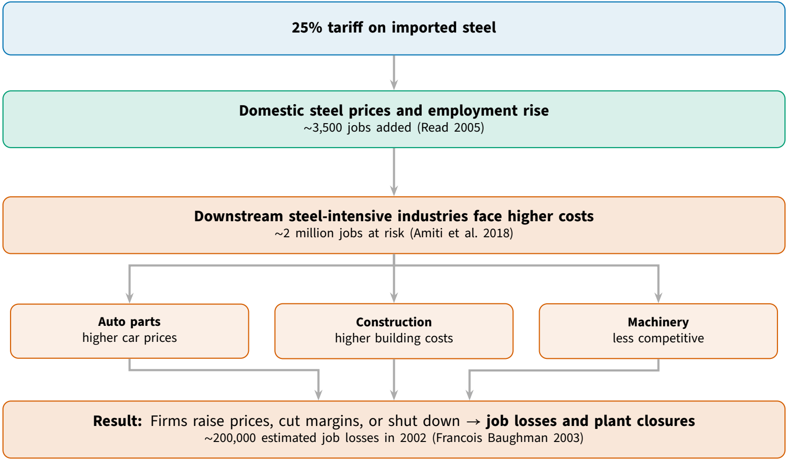

Steel Tariffs: What about Trade Wars?

> The video discussion did **not** account for trade wars!

Trade War

A cycle of retaliatory tariffs between countries, where each side raises trade barriers in response to the other's actions.

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WORLD • ASIA • CHINA

China Retaliates Against Trump Tariffs With Duties on American Meat and Fruit

Penalties range from 25% on American pork and eight other kinds of goods to 15% on fruit and 120 types of commodities

By Charles Hutzler
Updated April 1, 2018 10:47 pm ET

828 Gift unlocked article

When the U.S. raised tariffs, trading partners hit back.

American exporters—who had *nothing to do with steel*—paid the price.

Claim 3: Are Tariffs a Good Fix?

"If international trade is not good for every sector, tariffs can protect those affected sectors."

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"If international trade is not good for every sector, tariffs can protect those affected sectors."

Answer

It's complicated. Tariffs can protect specific *sectors*, but they raise costs for consumers and downstream firms. The costs often outweigh the benefits. Retaliation hurts unrelated industries.

There is no free lunch—every policy involves tradeoffs.

Outline

1. Is Trade Good for a Country?

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Let's Revisit Our Three Claims

1. "International trade is generally good for a country's economy."
   → **Yes**—comparative advantage grows the pie.

2. "International trade is generally good for every sector of a country's economy."
   → **No**—creates winners and losers. Transition is hard.

3. "Tariffs protect a country when international trade is bad for the economy."
   → **It's complicated**—protects some, taxes others. Costs often exceed benefits.